텍스트 분류 모델을 이용한 배달 앱 음식점 공지 오토 라벨링

이현민O 정형구*

서울시립대학교 컴퓨터과학부

yunoa64@gmail.com, hjung@uos.ac.kr

Auto Labeling Restaurant Notices of Delivery Applications using Text Classification Model

Hyunmin LeeO Hyunggu Jung*

Department of Computer Science and Engineering, University of Seoul

Abstract

Due to the outbreak of the COVID-19 pandemic and social distancing, the food delivery application market has shown rapid growth. This made customer reviews on restaurants of delivery applications influencing. However, fake reviews are being written to mislead customers with false information. Typically, review events held by restaurants encourage customers to leave positive reviews on the delivery applications, making it difficult for users to write objective and honest reviews. We try to solve this problem by creating a tool that labels restaurant notices promoting such review events. We first collected and preprocessed restaurant notice data and used the data to train and evaluate a classifier model.

1. Introduction

Due to the COVID-19 pandemic and social distancing, the delivery market is growing rapidly. Since delivery food transactions in Korea recorded KRW 1.1353 trillion in 2020 and KRW 1.8241 trillion in 2021, it is estimated that it will tentatively record KRW 2.2443 trillion in 2022. It is growing 97.7% compared to 2020 when COVID-19 gradually began to spread [1]. Delivery food orders using delivery applications account for 72.4% of the total delivery food market [2]. However, though restaurant income through delivery applications is greatly influenced by users' reviews and star ratings [2,3], it is reported that 15 to 30 percent of all online reviews were found to be faked [4]. Restaurant owners hold “review events” that ask consumers to write positive reviews in return for providing services, but such reviews tend to be false [5].

Prior studies created and evaluated systems that deal with online fake reviews such as preventing the alternation of reviews or detecting fake reviews written online. [6-9]. Other studies proposed an auto-labeling model for text classification from various data sources [10-14]. Nevertheless, to the best of our knowledge, no studies created and evaluated an auto-labeling model for classifying restaurants’ notices whether it holds review events. To fill this gap, we aim to answer the following research question: which classifier shows the best performance for labeling restaurant notices that hold review events?

*Corresponding author.

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. 2020R1G1A1009133).

To answer the question, we created and evaluated a tool that labels restaurant notices by checking whether it runs review events. Hence, our study makes the following contributions: (1) we present a way to collect and preprocess restaurant notice data used for training and evaluating a classification model; (2) we present the tool that automatically labels restaurants’ notices and gives information about review events using classifiers; (3) we present the performances of classifiers used for building a tool, and check which of them showed the greatest performance.

2. Related Work

Previous studies conducted research to develop and evaluate a system concerning online restaurant reviews. Saveetha and Maragatham [6] implemented a blockchain system that stores customer restaurant reviews so that they can’t be removed or altered. Anderson and Magruder [7] proposed a regression discontinuity design model and analyzed the positive effects of the high review ratings of Yelp. Wu et al. [8] trained and evaluated a Reliable Fake Review Detection (RFRD) framework using Yelp online reviews. Ko et al. [9] developed a model which classifies bloggers’ paid restaurant reviews.

Other researchers developed an auto labeling model and performed text classification. Xin et al. [10] used CNN and RNN classifiers to recognize certain entities on the given text and label a terrorism text. Hacioglu et al. [11] built an SVM model which labels text tokens with time expressions. Sovrano et al. [12] build a model to classify whether a paragraph of an official English resolution of the United Nations (UN) is related to one or more Sustainable Development Goals. Fazayeli et al. [13] built an auto labeling model which classifies git-issue-reports. Lin et al. [14] built a model which labels movie review comment text data

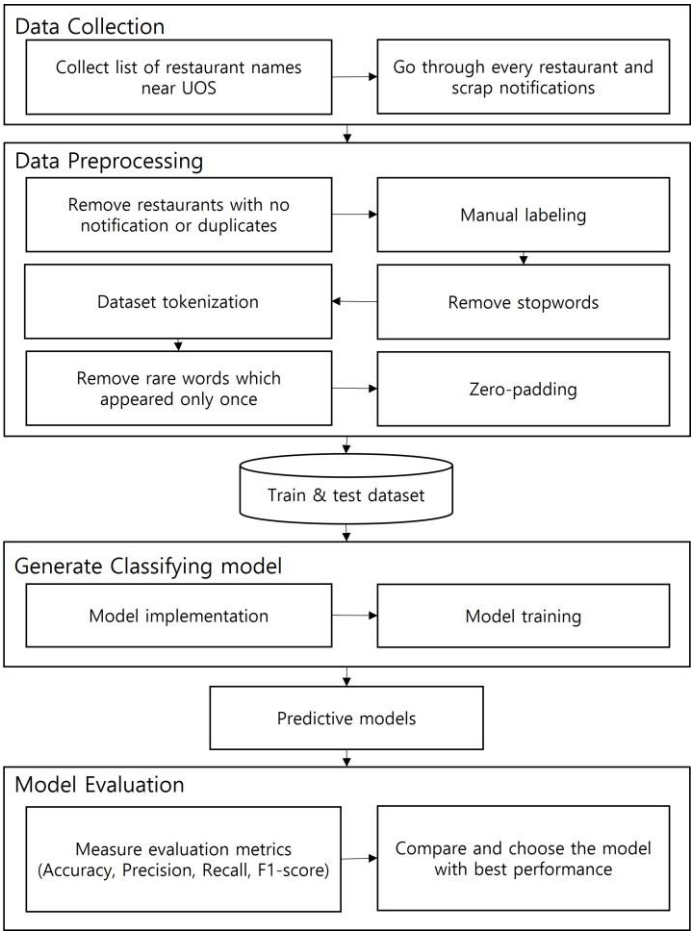

Figure 1. Overview of Tool Creating and Evaluating Process

as positive comments and negative comments. However, no studies focused on the restaurant reviews among online reviews. Also, none of those studies attempted to build a tool to automatically label restaurant notices in delivery applications to identify restaurants that hold review events.

3. Methods

In this section, we describe how we created our proposed tool. The overview of our process is shown in Figure 1. First, we collected restaurant notice text data from a delivery application Yogiyo[1]. Second, we obtained a dataset for classifier training and evaluation by the data preprocessing process. After that, we trained the proposed classifier with our preprocessed data.

3.1. Data Collection

To collect restaurants' notice text data from a delivery application, we scraped Yogiyo, using Selenium Chrome web driver. First, we collected a list of restaurant names near the University of Seoul (UOS) by setting our location to UOS and searching all available restaurants near UOS. We collected 2,677 restaurant names on Dec 31, 2021. Next, we searched these restaurants and scraped notice text data from Dec 29, 2021, to Dec 31, 2021.

[1] https://www.yogiyo.co.kr/

Table 1. Summary of the Test Data Results

	TP	FP	TN	FN
LSTM	167	11	119	4
CNN	171	20	110	0
RNN	161	23	107	10
SVM	113	88	42	58
KoBERT	144	72	58	27

Table 2. Summary of the Evaluation Metrics

	Accuracy	Precision	Recall	F1-score
LSTM	**0.9502**	**0.9382**	0.9766	**0.9570**
CNN	0.9334	0.8953	**1.0000**	0.9448
RNN	0.8903	0.8750	0.9415	0.9070
SVM	0.5150	0.5622	0.6608	0.6075
KoBERT	0.6711	0.6667	0.8421	0.7442

3.2. Data Preprocessing

We removed restaurant lists that did not contain any notices and removed duplicate notices. Then, we manually labeled the remaining notice data by judging whether we can see that a certain restaurant holds a review event. In our dataset, 628 notices were labeled as holding a review event, while the 574 notices were labeled as not a holding event.

To improve the performance of the classifier, we preprocessed the remaining notice text data. First, we removed every emoji and language other than Korean. Next, we removed stop words using the stop words list proposed by Kil [15]. Then, we tokenized the dataset into tokens and created a sequence vector using the Konlpy library [16]. After that, we excluded tokens that appeared only once on the entire dataset to reduce the curse of dimensionality. Finally, we used zero padding to unify each vector's length. However, this preprocessing method is not used when implementing the KoBERT model, because KoBERT is a pre-trained model and provides its own tokenizer function. As for the rest of the classifier (see Section 3.3), the data tokenization step is done as mentioned above.

After data preprocessing, we split the dataset into train and test datasets. We used 70% of the dataset for training, and the rest was used as a test dataset. Also, 20% of the training dataset was used for validation during each epoch.

3.3. Model Creation and Training

We chose KoBERT, LSTM, SVM, CNN, and RNN classifiers because they were used for text classifications in other domains [10, 11, 13, 14]. Since KoBERT offers a pre-trained model, we trained that model with our training dataset. As for the rest of the classifiers, we built our model using TensorFlow [17].

3.4. Model Testing

We evaluated our tool using a test dataset. During the test process, we calculated the models' performance using True-Positive (TP), False-Positive (FP), True-Negative (TN), and

False-Negative (FN). Positive stands for holding a review event, while negative stands for not holding an event. Table 1 shows our models' performances shown in the test dataset. Table 2 shows the values of each classifier's evaluation metrics.

4. Results

As we can see in Table 2, LSTM showed the best performance on Accuracy, Precision, and F1-score, while CNN showed the best performance on Recall. Though RNN did not reach LSTM and CNN, it also showed good performance. However, KoBERT tended to label test data as positive, as shown in Table 1, resulting in high Recall yet low Accuracy. And SVM showed the lowest in every metric. Therefore, LSTM and CNN showed high performance for classifying restaurants' notices because those two showed great performance on every evaluation metric.

5. Conclusions and Future Work

In this paper, we proposed a tool that labels restaurant notices promoting review events. The key contributions of our study are as follows: (1) we demonstrate how to collect and preprocess restaurants' notice data used for training and evaluating classifiers; (2) we present the auto-labeling tool that labels restaurants' notices about review events; (3) we present the performances of classifiers used for building the tool, and check which of them showed the greatest performance.

The limitations of our study are as follows: (1) our model does not consider images in the notice. While we were manually labeling our collected dataset, we noticed some restaurants place their review event information inside images, and not on texts; (2) we did not collect notice data from areas other than UOS, nor from other delivery applications; (3) we removed non-Korean texts during preprocessing before using the Konlpy tokenizer function, making us not able to use other languages and emojis; (4) we used only five classifiers to answer our research question.

Future studies remain to improve the performance of our proposed tool. First, we would need to consider images in restaurants' notices. That will allow us to find restaurants that place their review event information inside images. Second, we should collect more data from other delivery applications and restaurants in various places. Then, we may improve our model with various kinds of data, resulting in increased performance of the model. Third, we should consider the non-Korean text of restaurant notices because the other languages and emojis serve an important role. For example, sometimes "event" is written in English, and star emojis are used for appealing high star ratings. By considering those things, we may minimize information loss during preprocessing. Lastly, we would need to identify theoretical evidence for the classifiers we chose and use other classifiers to check whether there is a better one for our system. That will allow us to answer our RQ better. We hope our tool support researchers who aim to build a tool to detect fake reviews written for review events by showing the method to narrow the scope of the target restaurants.

References

[1] Korea, S. "Online Shopping." *Report material. (2022)*

[2] Heo, S. J., & Bae, H. J., "Analysis of the consumption pattern of delivery food according to food-related lifestyle," in *Journal of Nutrition and Health*, 53(5), 547-561, 2020.

[3] Lee, K. A., A study on measures to improve online consumer review systems. *Policy Analysis*, 16-23, 2016.

[4] Zhao, C., & Wang, C. A., "A cross-site comparison of online review manipulation using Benford's law," in *Electronic Commerce Research*, 1-42, 2021.

[5] Choi, S., "Consumers' propensity of post fake online reviews," 2015.

[6] Saveetha, D., & Maragatham, G., "Online Customer Reviews on Restaurant Using Blockchain.", 2021.

[7] Anderson, M., & Magruder, J, "Learning from the crowd: Regression discontinuity estimates of the effects of an online review database," *The Economic Journal*, *122*(563), 957-989, 2012.

[8] Wu, X., Dong, Y., Tao, J., Huang, C., & Chawla, N. V., "Reliable fake review detection via modeling temporal and behavioral patterns," In *2017 IEEE International Conference on Big Data (Big Data)* (pp. 494-499), IEEE, 2017

[9] Ko, M. C., Huang, H. H., & Chen, H. H., "Paid review and paid writer detection," In *Proceedings of the International Conference on Web Intelligence* (pp. 637-645), 2017.

[10] Xin, Z., Tianbo, W., Haiqiang, C., Qiang, Y., & Xiaohai, H., "Automatic Annotation of Text Classification Data Set in Specific Field Using Named Entity Recognition," in *2019 IEEE 19th International Conference on Communication Technology (ICCT)* (pp. 1403-1407), IEEE, 2019.

[11] Hacioglu, K., Chen, Y., & Douglas, B., "Automatic time expression labeling for english and chinese text," in *International Conference on Intelligent Text Processing and Computational Linguistics* (pp. 548-559). Springer, Berlin, Heidelberg. 2005.

[12] Sovrano, F., Palmirani, M., & Vitali, F., "Deep learning based multi-label text classification of UNGA resolutions," in *Proceedings of the 13th international conference on theory and practice of electronic governance* (pp. 686-695), 2020. - 10

[13] Fazayeli, H., Syed-Mohamad, S. M., & Akhir, N. S. M., "Towards auto-labelling issue reports for pull-based software development using text mining approach," in *Procedia Computer Science*, 161, 585-592, 2019.

[14] Lin, S., Zhang, R., Yu, Z., & Zhang, N., "Sentiment Analysis of Movie Reviews Based on Improved Word2vec and Ensemble Learning" in *Journal of Physics: Conference Series* (Vol. 1693, No. 1, p. 012088), 2020.

[15] Gill, H. H., "The Study of Korean Stopwords list for Textmining, URIMALGEUL: The Korean Language and Literature", 78, 1-25, 2018.

[16] Park, E. L., & Cho, S., "KoNLPy: Korean natural language processing in Python." in *Annual Conference on Human and Language Technology* (pp. 133-136). Human and Language Technology, 2014.

[17] Abadi, M. et al., "Tensorflow: Large-scale machine learning on heterogeneous distributed systems," in *arXiv preprint arXiv:1603.04467*, 2016.